Classical Field Theory — Exercise Sheet 23

A second KdV bracket and further conserved quantities

Solutions

Use Lecture 23's field space $\mathcal{S} = \mathcal{S}(\mathbb{R}, \mathbb{R})$, gradient space $\mathcal{B} = \mathbb{R} \oplus \mathcal{S}$ and admissible smooth functionals $\mathcal{A}_{\mathrm{KdV}}$. Write $D_x = \partial_x$ for spatial differentiation. Every time-dependent conclusion is conditional on a C^1 path in $\mathcal{S}$ satisfying KdV.

Exercise 23.1 (A variable Hamiltonian operator). For $u \in \mathcal{S}$ define the linear operator $J_2[u] : \mathcal{B} \to \mathcal{S}$ by

$$J_2[u]b = -b''' + 4ub' + 2u'b.$$

The prime means D_x, including when applied to a gradient. Define $\{F, G\}_2 = \int_{\mathbb{R}} F_u J_2[u] G_u \, \mathrm{d}x$.

(a) Prove skewness by integration by parts with both gradients in $\mathcal{B}$, keeping the endpoint primitive. Derive the gradient of $\{F, G\}_2$ and prove closure of the stated class.

(b) Prove Jacobi, also for $J_2 + \alpha D_x$ with every constant $\alpha \in \mathbb{R}$. Use the Hessian symmetry from Lecture 23 and account for the extra terms from differentiating the operator itself. A useful notation is $W(a, b) = ab' - a'b$.

(c) For $P = \frac{1}{2}\int u^2$ and $H = \int (u^3 + u_x^2/2)$, prove $J_2 P_u = D_x H_u$. Is the Gardner mass Casimir also a Casimir of this bracket?

The second KdV operator is discussed in Olver, *Applications of Lie Groups to Differential Equations*, second edition, Examples 7.6 and 7.9, pp. 438–439 and 444. The operator above states the course's scaling and signs explicitly; its proof is part of this problem.

Solution. Take $a, b \in \mathcal{B}$ for the skewness calculation. For the functional calculation below set $a = F_u$, $b = G_u$ and $H_F = D(F_u)(u)$, with $H_G = D(G_u)(u)$ defined in the same way. All terms in $J_2 b$ are Schwartz: differentiation removes the constant part of b, while the coefficients u, u' are Schwartz. Direct differentiation gives the skew primitive

$$aJ_2 b + bJ_2 a = D_x\big[-ab'' + a'b' - a''b + 4uab\big].$$

At each endpoint every displayed term tends to zero. The first three contain a derivative of a gradient, and the last contains u. Integrating proves skewness on $\mathcal{B}$.

Differentiate the bracket in a Schwartz direction h. The derivative of its operator, with b fixed, is $4hb' + 2h'b$. Hessian symmetry and skewness move the other two derivatives, giving

$$\begin{aligned} D\{F, G\}_2(u)h &= \int \big[(H_F h) J_2 b + a J_2(H_G h) + a(4hb' + 2h'b)\big]\mathrm{d}x \\ &= \int h\big[H_F J_2 b - H_G J_2 a + 2(ab' - a'b)\big]\mathrm{d}x. \end{aligned}$$

The last integration uses $4ab'h + 2abh' = 2(ab' - a'b)h + D_x(2abh)$, whose endpoint term vanishes. Thus the gradient is $H_F J_2 b - H_G J_2 a + 2W(a, b)$. It belongs to $\mathcal{B}$ because both Hessian directions $J_2 b, J_2 a$ belong to $\mathcal{S}$, and $W(a, b)$ is Schwartz. The displayed formula is smooth by the same composition and product calculus as Lecture 23. It proves bracket closure.

For Jacobi write also $c = K_u$. Substitution of this gradient in the cyclic sum $\{\{F, G\}_2, K\}_2 + \{\{G, K\}_2, F\}_2 + \{\{K, F\}_2, G\}_2$ cancels the six Hessian terms in pairs by symmetry. Every

Hessian direction is J_2a, J_2b or J_2c in $\mathcal{S}$. The remaining sum is

$$2\int \big[W(a,b)J_2c + W(b,c)J_2a + W(c,a)J_2b\big]\mathrm{d}x.$$

The coefficients multiplying u and u' vanish pointwise:

$$W(a,b)c' + W(b,c)a' + W(c,a)b' = 0, \qquad W(a,b)c + W(b,c)a + W(c,a)b = 0.$$

Expanding the six products proves each equality. For the third-derivative part put $T = W(a,b)c'' + W(b,c)a'' + W(c,a)b''$. Its derivative is

$$T' = W(a,b)c''' + W(b,c)a''' + W(c,a)b'''.$$

Indeed, the extra terms are the cyclic sum $(ab'' - a''b)c''$, which cancels in pairs. Each term of T contains derivatives of gradients and tends to zero at both ends. The remaining Jacobi sum is therefore $-2\int T' = 0$. Bilinearity, skewness and the product rule follow from the definition and the usual product gradient. This proves the Poisson assertion. Adding αD_x preserves skewness and closure. In the remaining Jacobi expression its extra contribution is $2\alpha\int[W(a,b)c' + W(b,c)a' + W(c,a)b'] = 0$. Thus every $J_2 + \alpha D_x$ is Poisson: the two brackets are compatible.

Finally, $P_u = u$ and $H_u = 3u^2 - u''$ give

$$J_2P_u = -u''' + 4uu' + 2u'u = 6uu' - u''' = D_xH_u.$$

Thus $u_t = J_2P_u$ is the same KdV equation as $u_t = D_xH_u$. The mass gradient is one, and $J_21 = 2u'$ is generally nonzero. Fix a nonconstant profile u_* and the smearing $g = u_*'$. At that field the linear observable $u(g)$ detects it: $\{u(g), M\}_2[u_*] = 2\int (u_*')^2 > 0$. The Gardner and second-operator degeneracies differ.

Exercise 23.2 (Finding a higher polynomial invariant). For real constants A, B, C put

$$I_{A,B,C}[u] = \int_{\mathbb{R}} \big(Au^4 + Bu(u')^2 + C(u'')^2\big)\mathrm{d}x.$$

Determine all triples such that $DI_{A,B,C}(u)(6uu' - u''') = 0$ for every $u \in \mathcal{S}$. Conclude conservation along every stated Schwartz KdV path. Show necessity as well as sufficiency; do not infer independence of two integrals merely by looking at different integrands.

Solution. Integration by parts gives

$$(I_{A,B,C})_u = 4Au^3 - B(u')^2 - 2Buu'' + 2Cu''''.$$

Pair this gradient with $6uu' - u'''$. Terms which are single total derivatives have zero integral. For the others use

$$\int u^2u'u'' = -\int u(u')^3, \qquad \int (u')^2u''' = -2\int u'(u'')^2, \qquad \int uu''u''' = -\frac{1}{2}\int u'(u'')^2.$$

These are derivatives of $u^2(u')^2$, $(u')^2u''$ and $u(u'')^2$. They also imply $\int u^3u''' = 3\int u(u')^3$ and $\int uu'u'''' = \frac{5}{2}\int u'(u'')^2$. Consequently the complete directional derivative is

$$DI_{A,B,C}(u)(6uu' - u''') = (6B - 12A)\int u(u')^3 + (30C - 3B)\int u'(u'')^2.$$

This proves sufficiency if $B = 2A$ and $B = 10C$.

For necessity let $u_* = (1 + x)e^{-x^2}$. Gaussian integration gives

$$\int u_*(u_*')^3 = \frac{\sqrt{\pi}}{8}, \qquad \int u_*'(u_*'')^2 = \frac{32\sqrt{3\pi}}{81}.$$

Both are nonzero. Replace u_* by au_* for arbitrary real a. The first integral scales as a^4 and the second as a^3. A polynomial in a vanishes identically only when both coefficients vanish. Thus $B = 2A$ and $B = 10C$, equivalently $(A, B, C) = C(5, 10, 1)$, including the zero triple. The chain rule along a $C^1(\mathcal{S})$ solution now proves conservation. No existence theorem for arbitrary initial data was used.

Exercise 23.3 (Classifying Gardner Casimirs). A Casimir of Lecture 23's Gardner bracket is an admissible functional commuting with every admissible functional. Prove that these are exactly

$$F[u] = \psi(M[u]), \qquad M[u] = \int_{\mathbb{R}} u \, \mathrm{d}x,$$

for smooth $\psi : \mathbb{R} \to \mathbb{R}$. Constant scalar functionals are included. Explain why this does not say that every local polynomial density is a nonlinear function of the density u. Compare the local-functional discussion in Olver, Example 7.13, p. 446: the present admissible class also allows nonlocal composites $\psi(M)$.

Solution. Let $a = F_u[u] \in \mathcal{B}$ at a fixed field. For every Schwartz g, the Casimir property gives

$$0 = \{F, u(g)\} = \int ag' = -\int a'g.$$

Here a' is Schwartz. The compact-test fundamental lemma makes $a' = 0$, so a is spatially constant. Its value may depend on the entire field.

If u and v have the same mass, the straight segment $u_s = u + s(v - u)$ stays in $\mathcal{S}$ and has that mass. The chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}s}F[u_s] = \int F_u[u_s](v - u) = 0,$$

because the gradient is spatially constant and $\int (v - u) = 0$. Thus F is constant on every affine mass fibre. Choose $\eta \in \mathcal{S}$ with integral one and set $\psi(s) = F[s\eta]$. It is smooth by composition, and u and $M[u]\eta$ lie in the same fibre. Hence $F[u] = \psi(M[u])$.

Conversely, this composite has gradient $\psi'(M[u])$, a constant element of $\mathcal{B}$ depending smoothly on u. It is admissible, and for any admissible G,

$$\{F, G\} = \psi'(M[u]) \int D_x G_u[u] \, \mathrm{d}x = 0.$$

The integral vanishes because the gradient's derivative is Schwartz and its constant part has the same limit at both ends. This includes constant ψ. For example $M[u]^2$ is such a functional, whereas $\int u(x)^2 \mathrm{d}x$ is the momentum functional and has nonzero gradient derivative for a nonconstant field. Nonlinear composition with the integrated mass is different from integrating a nonlinear density.